

DAM
CSEC-409

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(7th term)

UOS

SQL * Plus

Definitions

SQL * Plus is a command-line interface tool provided by Oracle for interacting with Oracle databases.

- o It allows users to:
- o Execute SQL statements
- o manage database objects
- & Perform various tasks directly from the command prompt or terminal.

Key features of SQL * Plus:

Connectivity:

- o SQL * plus allows connections to Oracle databases using credentials and service names.
- o Supports both local and remote connections.

Execution:

- o Execute SQL statements and PL/SQL blocks directly from the command prompt.

- Provide immediate results.

Scriptings

- can create and execute script containing multiple SQL commands.
- Scripts are stored as text files and executed through SQLPlus CLI.

Output formatting

- it offers customize display of query results with options for column widths, alignments and pagination.

Spoolings

- Redirect SQL output to files for saving query results or generating reports in various formats.

Variable substitutions

- Use variables in SQL statements for enhanced flexibility and reusability of scripts.

Database administration

- SQL * PLUS provides administrative commands for managing database objects, users, roles, and privileges, facilitating tasks like table creation, data modification and security management.

Error handling and logging

- it captures and display errors that occur during SQL execution and offers logging capabilities for recording executed commands, queries and errors for analysis.

Customization

- Define environment variables, set display preferences, and create custom scripts.

Integrations

- It can be integrated with shell scripts, batch files and other programming languages for automation, enabling the incorporation of SQL * PLUS functionality into larger workflows.

History of SQL Plus:

- The author of SQL * Plus was Bruce Scott.

- The first version of SQL * Plus was called UI ("User friendly interface").

SQL VS SQL*Plus:

- SQL is a language.
- SQL * Plus is a tool to run SQL and PL/SQL.
- it is character based interactive tool, that runs in a GUI environment.

Q-Explain tasks of an Oracle DBA?

⇒ The approach for designing, implementing and maintaining an Oracle database involves the following tasks:

- ⇒ Evaluating the data server hardware
- ⇒ Installing the Oracle Software.
- ⇒ Planning the database and security strategy.
- ⇒ Creating, migrating and opening the databases.
- ⇒ Backing up the database
- ⇒ Enrolling system users and planning for their Oracle Network access.
- ⇒ Implementing the database design.
- ⇒ Recovering from database failure
- ⇒ Monitoring database performance.

Tasks of an DBA:

- The DBA is typically responsible for installing the Oracle software and creating the database.
- Also responsible for creating database storage structures, such as tablespaces.
- Also create schema or set of objects to hold application data.

- Also ensure that the database is typically available for users.
- Responsible for backing-up the database on a regular basis and monitoring the performance of the database.

Q- SQL Plus vs SQL

SQL Plus	SQL
○ Tool	○ language
○ Invented by Oracle Corporation	○ was invented by IBM
○ it will act as a editors for writing and executing SQL query.	○ It will provide sub-language like DDL, DML etc

